

Robot-Centric Driving Practice

Two-Hour Full-Field Session

Purpose

Build driver confidence with robot-centric mecanum controls, improve precise alignment, and develop repeatable full-field cycles under match pressure.

In robot-centric control, joystick directions are relative to the robot: pushing forward always moves the robot toward its current front, regardless of where it is facing on the field.

Equipment and Setup

- Competition robot, charged batteries, Driver Station, and gamepads
- Full field with normal obstacles and scoring structures
- 8–12 cones, game pieces, or floor markers
- Tape for lanes and stopping boxes; stopwatch and score sheet

Inspect the robot, verify controls and speed modes, and agree on an emergency-stop call. Nobody enters the field until the robot is disabled.

Session Schedule

Time	Exercise	Primary focus
0:00–0:10	Setup and controls check	Safety and consistent response
0:10–0:25	Orientation warm-up	Robot-centric direction awareness
0:25–0:45	Precision lanes	Straight driving and strafing
0:45–1:05	Slalom and rotation	Translation while controlling heading
1:05–1:25	Target approach drill	Accurate mechanism positioning
1:25–1:45	Full-field routes	Speed and repeatability
1:45–1:57	Timed match cycles	Performance under pressure
1:57–2:00	Review	Choose the next practice priority

0:00–0:10 — Setup and Controls Check

1. Inspect wheels, hubs, mechanisms, battery, and wire routing.
2. Clear the field and mark corners, center, and scoring/loading positions.
3. Confirm joystick mapping, drive direction, speed modes, and emergency stop.
4. Test forward, backward, both strafes, and rotation.
5. Each driver completes one slow perimeter lap.

Do not continue until the robot responds predictably and the driver can stop it safely.

0:10–0:25 — Orientation Warm-Up

1. From center, face the audience side and drive to each wall using translation only.
2. Rotate 90 degrees and repeat; then repeat from 180 and 270 degrees.
3. The coach calls a field location; the driver selects the correct robot-centric input.
4. Reset to a different heading after every few calls.

Success target: 10 correct initial movements in a row.

0:25–0:45 — Precision Lanes

Create two parallel lanes and a stopping box at each end.

1. Drive forward through a lane, stop in the box, and reverse back.
2. Strafe left and right through the second lane without rotating.
3. Drive a square: forward, right strafe, backward, left strafe.
4. Repeat at slow, medium, and full speed.
5. Narrow the lanes after three clean runs.

Record time, cones touched, and whether the robot finishes inside the stopping box. Prioritize a clean path before increasing speed.

0:45–1:05 — Slalom and Rotation

Place five or six cones across the field.

1. Slalom without rotating the robot.
2. Repeat while keeping the robot pointed toward one wall.
3. Repeat with a 90-degree turn at every other cone.
4. Drive a figure eight around two widely spaced markers.
5. Run the course forward and in reverse.

Success target: three consecutive clean runs before increasing speed.

1:05–1:25 — Target Approach Drill

Mark several realistic scoring, pickup, or interaction positions. For each target:

1. Start from center or a realistic collection area.
2. Approach and stop in scoring alignment.
3. Complete the pickup or scoring action when possible.
4. Back away cleanly without contacting the target.
5. Repeat from a different robot heading.

Practice forward, backward, sideways, and diagonal approaches. Allow one controlled final correction after a diagonal approach.

Success target: 8 successful alignments out of 10 attempts.

1:25–1:45 — Full-Field Routes

Start → pickup area → scoring area → far corner → scoring area → park

1. Learn the route at 50% speed and repeat until clean.
2. Increase speed without sacrificing control.
3. Alternate clockwise and counterclockwise routes.
4. Change the starting orientation between runs.

Penalty	Added time
Cone or field contact	+2 seconds

Missed stopping or scoring position	+3 seconds
Wrong initial direction or major correction	+2 seconds

1:45–1:57 — Timed Match Cycles

Run three rounds of three minutes driving plus one minute of feedback or driver change. Give match-style objectives rather than a memorized path—for example, collect near-side, score twice, travel far-side, then park. Use only normal match communication. Record cycles, collisions, missed alignments, and orientation confusion.

1:57–2:00 — Review

Each driver names one improvement, one recurring mistake, and one exercise to repeat. Record the fastest clean route, average cycles, alignment success, and collisions as the baseline.

Coaching Priorities

1. **Accuracy before speed:** clean routes beat rushed routes with corrections.
2. **Look ahead:** focus on the destination and field traffic.
3. **Use deliberate headings:** choose a useful orientation before approaching.
4. **Make small final inputs:** reduce speed near targets.
5. **Reset after mistakes:** stop, locate the robot's front, and choose deliberately.

Practice Scorecard

Metric	Baseline	Best	Next target
Precision-lane clean runs			
Target alignments out of 10			
Fastest clean full-field route			
Average cycles per timed round			
Collisions during timed rounds			
Wrong-direction starts			